

$$0.4 - 0$$

4. Answer: D

Plot a graph of v against t . The graph will have a constant gradient g .

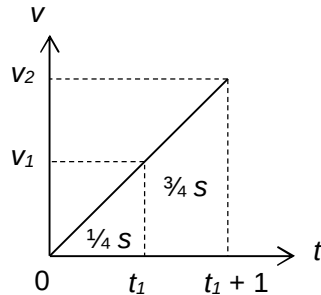
$$v_1 = gt_1 \dots\dots\dots (1)$$

$$v_2 = g(t_1 + 1) \dots\dots\dots (2)$$

Since the distance covered in the last second is $\frac{3}{4}$ of the whole height s , by comparing the areas under the graph,

$$\frac{\frac{1}{2}v_1t_1}{\frac{1}{2}v_2(t_1 + 1)} = \frac{1}{4} \dots\dots\dots (3)$$

Substituting (1) and (2) into (3),



$$\frac{gt_1^2}{g(t_1 + 1)^2} = \frac{1}{4}$$

$$\Rightarrow 3t_1^2 - 2t_1 - 1 = 0$$

$$\therefore t_1 = 1.0 \text{ s}$$

$$\text{Thus, total time} = t_1 + 1 = 2.0 \text{ s}$$

5. Answer: D